



TEST CODE **02238020**

FORM TP 2014237

MAY/JUNE 2014

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®

PHYSICS

UNIT 2 – Paper 02

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SIX questions in TWO sections. Answer ALL questions.
2. For Section A, write your answers in the spaces provided in this booklet.
3. For Section B, write your answers in the spaces provided at the end of each question in this booklet.
4. All working MUST be shown.
5. You may use a silent, non-programmable calculator to answer questions, but you should note that the use of an inappropriate number of figures in answers will be penalized.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

Copyright © 2012 Caribbean Examinations Council
All rights reserved.

02238020/CAPE 2014

LIST OF PHYSICAL CONSTANTS

Speed of light in free space	c	=	$3.00 \times 10^8 \text{ m s}^{-1}$
Permeability of free space	μ_0	=	$4\pi \times 10^{-7} \text{ H m}^{-1}$
Permittivity of free space	ϵ_0	=	$8.85 \times 10^{-12} \text{ F m}^{-1}$
			$\frac{1}{4\pi \epsilon_0} = 9.0 \times 10^9 \text{ m F}^{-1}$
Elementary charge	e	=	$1.60 \times 10^{-19} \text{ C}$
Planck's constant	h	=	$6.63 \times 10^{-34} \text{ J s}$
Unified atomic mass constant	u	=	$1.66 \times 10^{-27} \text{ kg}$
(Energy equivalence)	$1u$	=	$931 \text{ MeV}/c^2$
Rest mass of electron	m_e	=	$9.11 \times 10^{-31} \text{ kg}$
Rest mass of proton	m_p	=	$1.67 \times 10^{-27} \text{ kg}$
Acceleration due to gravity	g	=	9.81 m s^{-2}
1 Atmosphere	Atm	=	$1.00 \times 10^5 \text{ N m}^{-2}$
Avogadro's constant	N_A	=	$6.02 \times 10^{23} \text{ per mole}$

GO ON TO THE NEXT PAGE

SECTION A

Answer ALL questions.

Write your answers in the spaces provided in this booklet.

- 1.** (a) (i) Define the term 'capacitance'.

[1 mark]

- (ii) State the unit in which capacitance is measured.

[1 mark]

- (iii) Identify the two SI units that comprise the unit for capacitance stated in (a) (ii) above.

[1 mark]

- (b) Figure 1 shows a circuit including two capacitors in series, where $C_B = 60 \mu\text{F}$.

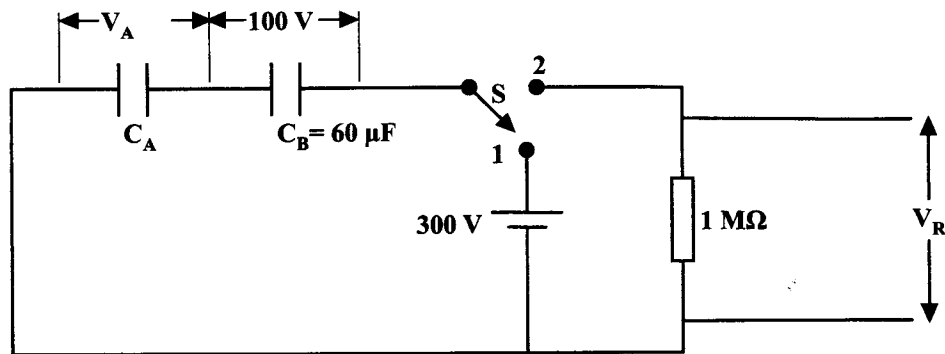


Figure 1

- (i) Calculate the value of the voltage, V_A , when the switch, S , is in Position 1.

[2 marks]

- (ii) Use your results from (b) (i) above to calculate the value of the capacitance, C_A .

[3 marks]

GO ON TO THE NEXT PAGE

- (iii) Show that the two capacitors in series in Figure 1 are equivalent to a single capacitor of value $20\ \mu\text{F}$.

[2 marks]

- (c) The switch, S, is placed in Position 1 for several minutes and then changed to Position 2 and, V_R , the voltage across the resistor recorded every 20 seconds. The results obtained are presented in Table 1.

TABLE 1

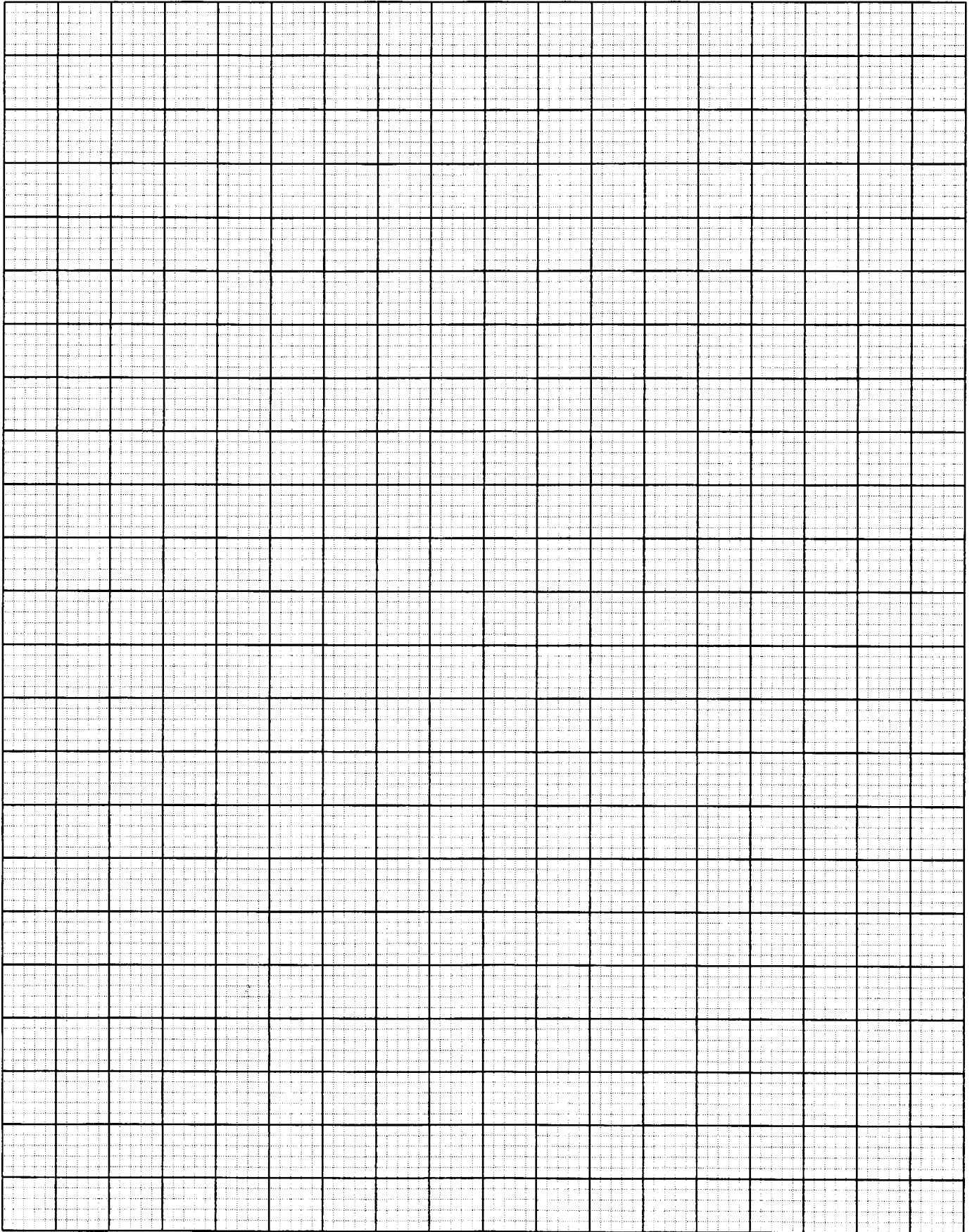
Time t (s)	Voltage $\times 10^2$ V_R (V)
0	3.00
20	1.11
40	0.42
60	0.15
80	0.06

- (i) On the grid provided on page 7, plot a graph of V_R against t for these results. **[3 marks]**
- (ii) Draw the tangent to the curve at the point where $t = 0$. **[1 mark]**
- (iii) Hence, determine the time constant of the voltage decay.

[1 mark]

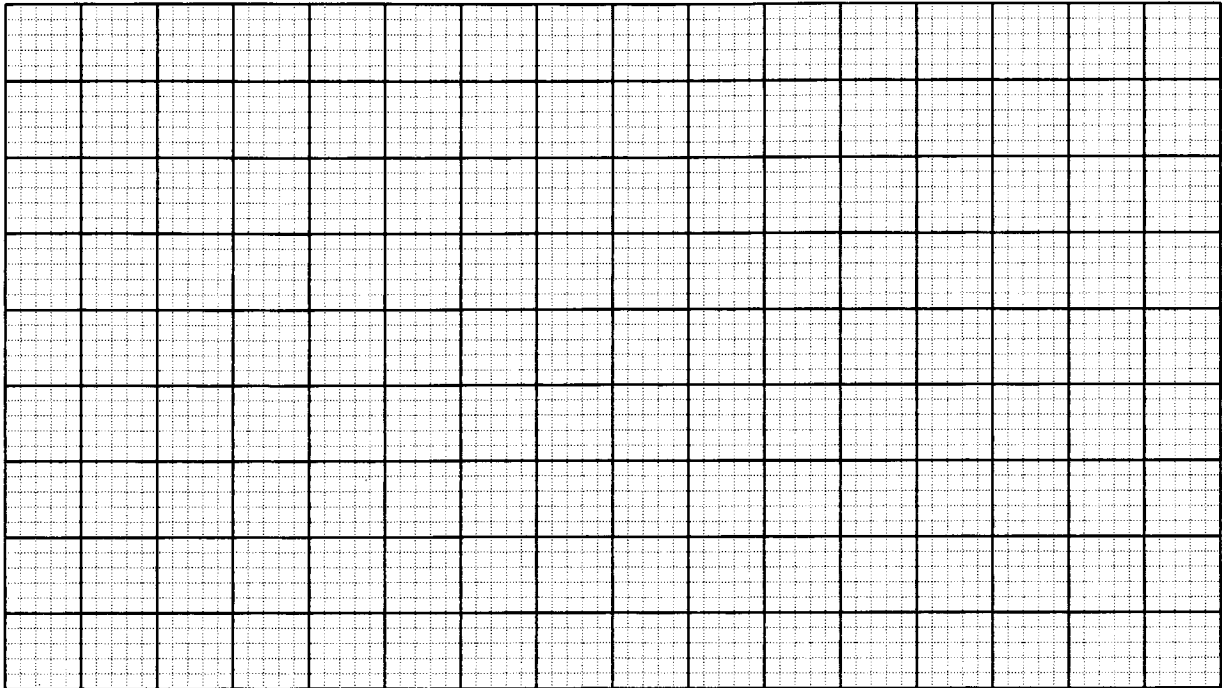
Total 15 marks

GO ON TO THE NEXT PAGE



NOTHING HAS BEEN OMITTED.

2. (a) On the grid provided below, draw a fully labelled sketch of the reverse bias I–V characteristic of a typical silicon p–n junction diode.



[3 marks]

- (b) The forward bias characteristic I_D vs V_D of a junction diode may be modelled by the approximate Shockley equation

$$I_D = I_S \exp (V_D/V_T), \text{ where}$$

I_D = diode forward current

I_S = diode reverse saturation current

V_D = diode forward biased volt drop

V_T = thermal volt drop = 27 mV (at 30°C).

The results shown in Columns 1 and 2 of Table 2 were obtained in an experimental determination of the forward bias characteristic of a diode.

TABLE 2

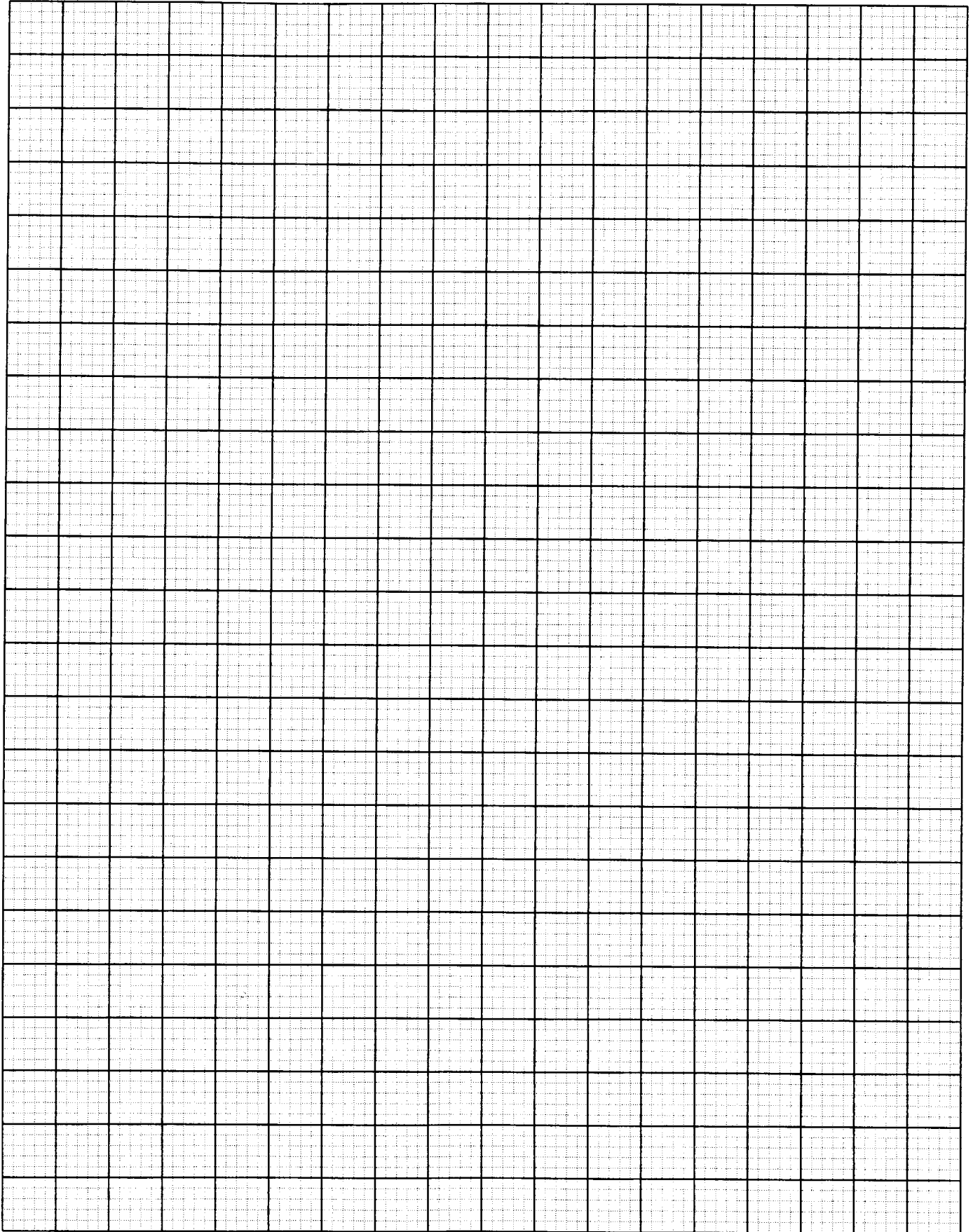
V_D/V_T	$I_D(\text{mA})$	$\ln(I_D)$
10	0.55	
12	4.08	
13	11.1	
14	30.2	
16	223	

- (i) Complete Column 3 of Table 2. **[3 marks]**
- (ii) On the grid provided on page 11, draw a graph of $\ln(I_D)$ vs V_D/V_T . **[5 marks]**
- (iii) Use your graph to calculate the reverse saturation current for this diode.

[4 marks]

Total 15 marks

GO ON TO THE NEXT PAGE



GO ON TO THE NEXT PAGE

3. (a) Write a brief description of an experiment to determine the half-life of a radioactive sample.

[5 marks]

- (b) State THREE properties of a radioisotope which would make it suitable for use in radiotherapy.

[3 marks]

GO ON TO THE NEXT PAGE

- (c) Carbon-14, $^{14}_6\text{C}$, has been used to date archaeological samples. It is produced naturally in the upper atmosphere by neutron bombardment of atmospheric nitrogen. Given that atmospheric nitrogen is $^{14}_7\text{N}$, derive the nuclear equation for the natural production of carbon-14.

[2 marks]

- (d) Carbon-14 is unstable and decays to nitrogen by beta emission. This process has a half-life of 5730 years. Determine the percentage of carbon-14, relative to the initial amount, which would remain after 20 000 years.

(5 marks)

Total 15 marks

GO ON TO THE NEXT PAGE

SECTION B

Answer ALL questions.

Write your answers in the spaces provided at the end of each question.

4. (a) Define the term 'electric field strength' and state the unit in which it is measured. **[3 marks]**
- (b) Two horizontal metal plates, in vacuo, are separated by 1 cm and a potential difference of 1 kV applied between them as shown in Figure 2.

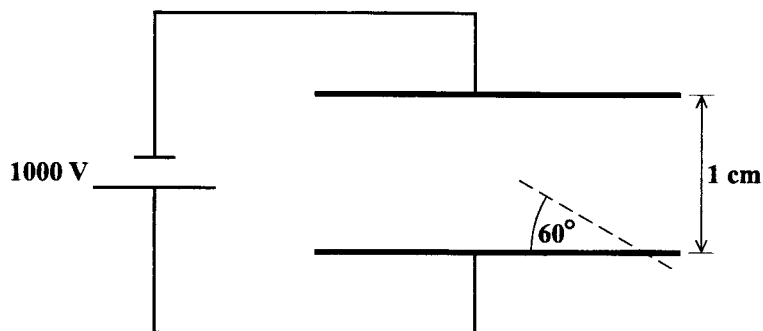


Figure 2

- Calculate the electric field strength between the plates. **[2 marks]**
- (c) A beam of electrons is projected through a hole in this positive plate with velocity, v , at an angle of 60° to the plate.
- (i) Describe the subsequent motion of the electron beam. **[1 mark]**
- (ii) Compare the motion of the electrons in the electric field with that of an object in the earth's gravitational field by considering the effect of the fields on the horizontal and vertical velocity components. **[4 marks]**
- (d) What must be the value of v if the beam is to just touch the negative plate? **[5 marks]**

Total 15 marks

GO ON TO THE NEXT PAGE

Write the answer to Question 4 here.

(a)

(b)

(c)

(i)

Write the answer to Question 4 here.

(ii)

(d)

GO ON TO THE NEXT PAGE

5. (a) Draw truth tables and the circuit symbols for the 2-input NAND gate and the XNOR gate. [4 marks]
- (b) The NAND gate and the NOR gate are sometimes referred to as universal gates because all other logic gates can be constructed from combinations of any one of them. Draw the NOR equivalents of the
- (i) AND gate [2 marks]
 - (ii) OR gate [1 mark]
 - (iii) NAND gate. [1 mark]
- (c) (i) Show, by using a truth table, that the circuit in Figure 3 produces the Exclusive NOR operation. [3 marks]

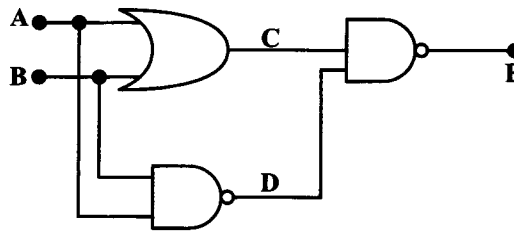


Figure 3

- (ii) Hence, develop a circuit using NOR gates **only** which will provide Exclusive NOR operation. [4 marks]

Total 15 marks

Write the answer to Question 5 here.

(a)

GO ON TO THE NEXT PAGE

Write the answer to Question 5 here.

(b) (i)

(ii)

(iii)

GO ON TO THE NEXT PAGE

Write the answer to Question 5 here.

(c) (i)

(ii)

6. (a) With the aid of a CLEARLY LABELLED diagram, describe the way in which x-rays are usually produced for use in diagnostic medicine and dentistry. **[8 marks]**
- (b) Express the energies of x-ray photons of wavelength 10^{-11} metres in electron Volts. **[3 marks]**
- (c) The absorption coefficient of x-rays in a given material is dependent on the energies of the incident photons. Consider the case of a hypothetical material with an absorption coefficient of 0.3 cm^{-1} at the wavelength used in 6 (b).
- What thickness of the material would be required to attenuate the intensity of a beam of x-rays to 1% of the incident intensity? **[4 marks]**

Total 15 marks

Write the answer to Question 6 here.

(a)

GO ON TO THE NEXT PAGE

Write the answer to Question 6 here.

(b)

(c)

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.